

Density

2026-09-27

Definition 0.1. For $A \subseteq \Gamma$ and Følner sequence $\Phi = (\Phi_N)_{n=1}^\infty$, we write

$$\begin{aligned}\bar{d}_\Phi(A) &= \limsup_{N \rightarrow \infty} \frac{|A \cap \Phi_N|}{|\Phi_N|} \\ \underline{d}_\Phi(A) &= \liminf_{N \rightarrow \infty} \frac{|A \cap \Phi_N|}{|\Phi_N|}\end{aligned}$$

to be the *upper and lower densities of A with respect to Φ* , respectively. If these agree, then we can write

$$d_\Phi(A) = \lim_{N \rightarrow \infty} \frac{|A \cap \Phi_N|}{|\Phi_N|}$$

to be the *density of A with respect to Φ* . We also define the *upper Banach density of A* by

$$\bar{d}(A) = \sup\{d_\Phi(A) : \text{for Følner sequences } \Phi \text{ where } d_\Phi(A) \text{ exists}\}.$$

Proposition 0.1 (Monotonicity). *Let $A, B \subseteq M$ be subsets. For $A \subseteq B$, we have*

$$\bar{d}_\Phi(A) \leq \bar{d}_\Phi(B),$$

and

$$\underline{d}_\Phi(A) \leq \underline{d}_\Phi(B).$$

Proposition 0.2 (Translation Invariance). *Let $B \subseteq M$ be a subset. For all $m \in M$, we have that*

$$\bar{d}_\Phi(B) = \bar{d}_{m \cdot \Phi}(B) = \bar{d}_\Phi(m \cdot B)$$

and

$$\underline{d}_\Phi(B) = \underline{d}_{m \cdot \Phi}(B) = \underline{d}_\Phi(m \cdot B).$$

Lemma 0.1 (Upper/Lower Density Pairwise Additive). *Let $A, B \subseteq M$ be subsets such that $d_\Phi(A)$ and $d_\Phi(B)$ both exist, then we have*

$$\begin{aligned}\bar{d}_\Phi(A \cup B) + \underline{d}_\Phi(A \cap B) &= d_\Phi(A) + d_\Phi(B), \\ \underline{d}_\Phi(A \cup B) + \bar{d}_\Phi(A \cap B) &= d_\Phi(A) + d_\Phi(B).\end{aligned}$$

Further, we have $d_\Phi(A \cap B)$ exists if and only if $d_\Phi(A \cup B)$ exists.